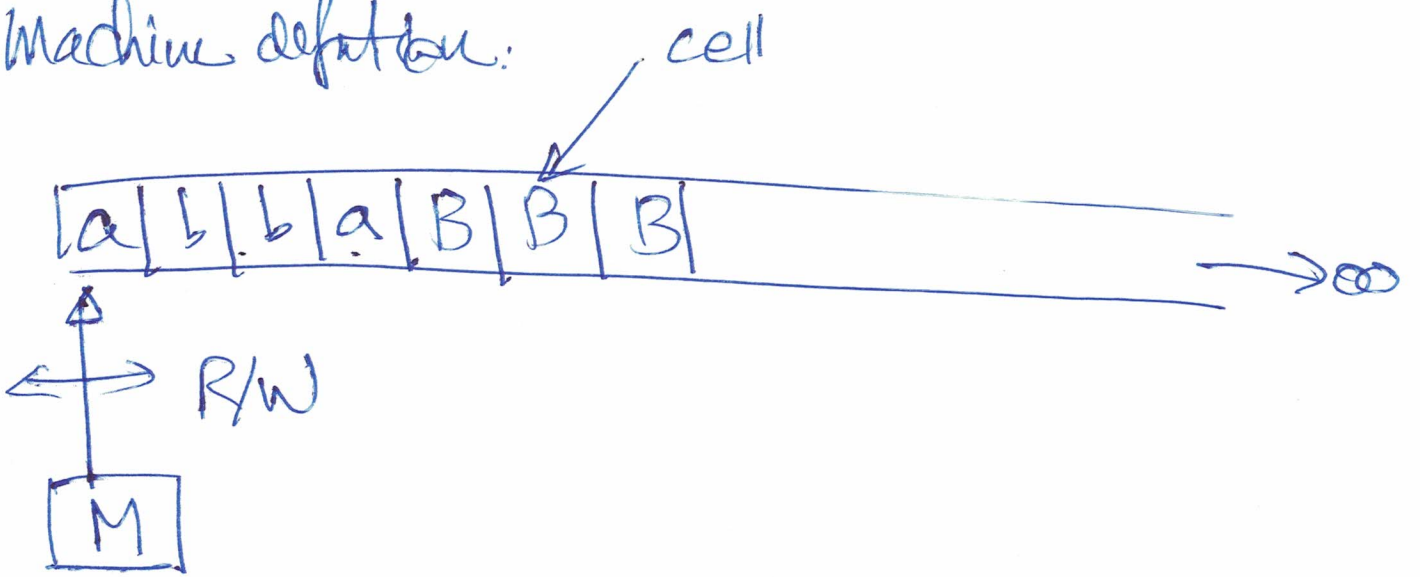
Cpts 317. 11/16.

Turing machine definition:



A TM M is a tuple:

$$\langle Q, \Sigma, \Gamma, q_0, \delta, B, F \rangle$$

where: Q: control states.

Σ: input alphabet,

Γ: tape alphabet; with $\Sigma \subseteq \Gamma$.

$q_0 \in Q$: initial state.

$F \subseteq Q$: accepting states;

$B \in \Gamma$: blank symbol.

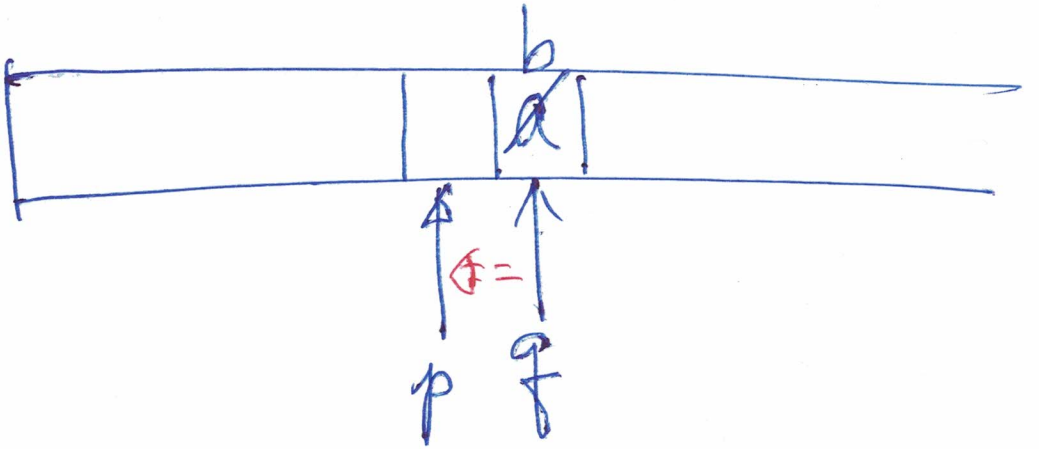
The instructions are def'ed in δ:

Format:

left.

$$\delta(q, a) = (p, b, L)$$

meaning: M is now at state q, reading a, Next, M switches to state p, change a into b, moves to left.

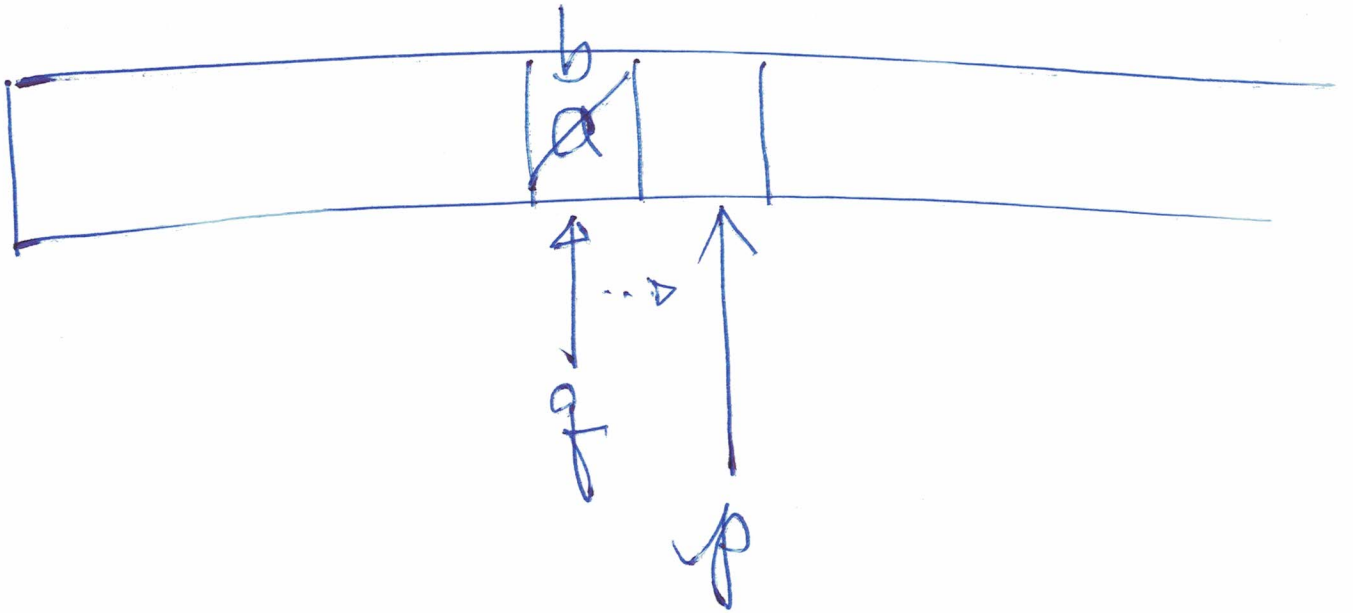


Format:

Right

$$\delta(q, a) = (p, b, R)$$

meaning: M is now at state q, reading a. Next M switches to state p, changes a to b, moves to the right.

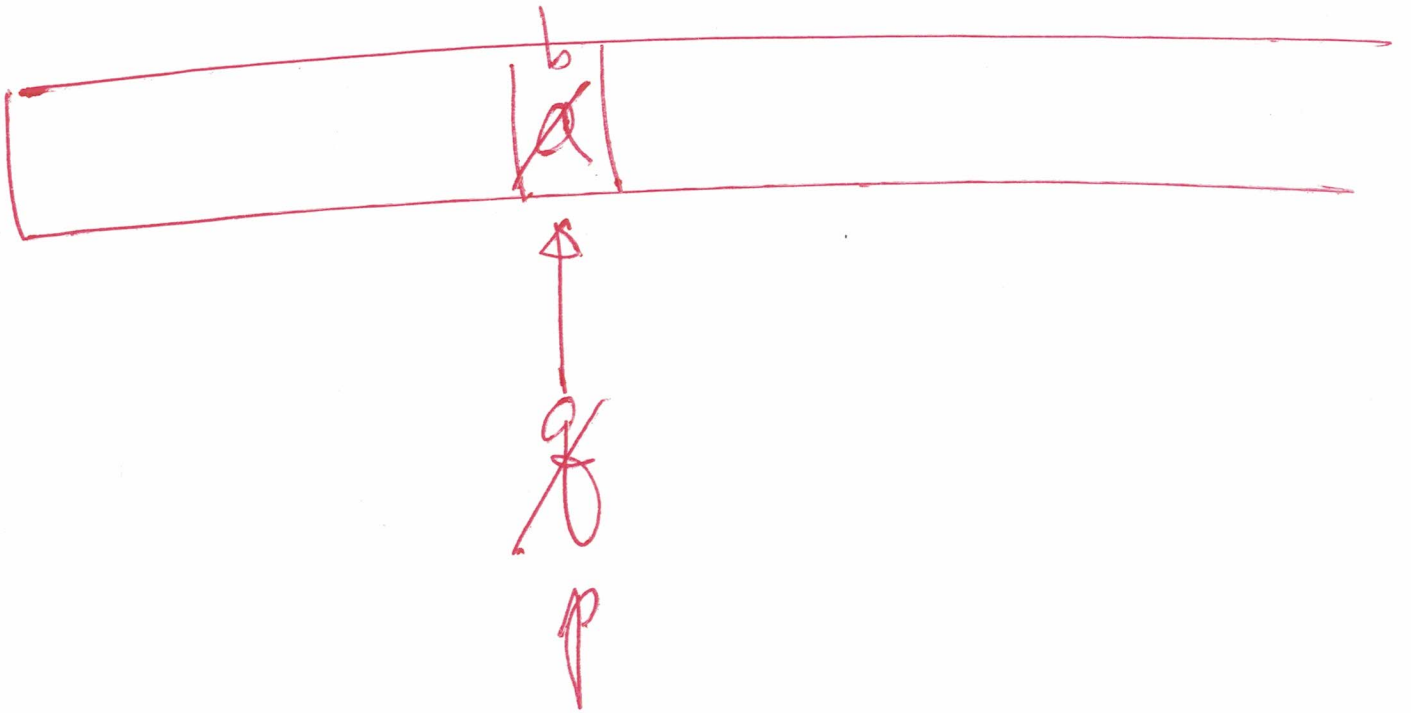


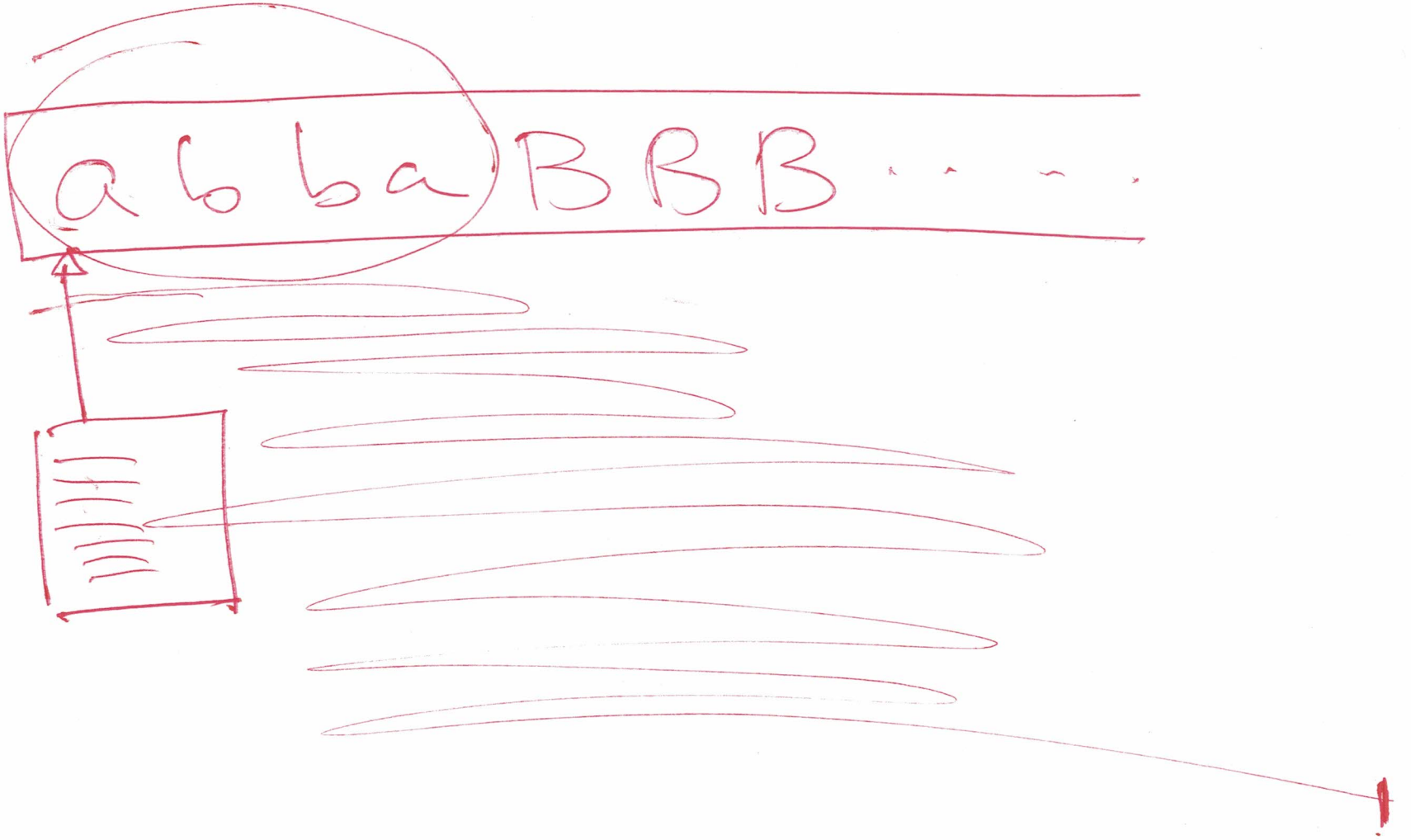
Format:

stay

$$\delta(q, a) = (p, b, S)$$

meaning: M is now at state q, reading a. Next, M switches to state p, changes a to b, M's head doesn't move.




abbaBBB...

Definitions.

When TM M runs on input w, if M eventually enters an accepting state, then we say that M accepts w (M says yes on w).

Define: $L(M) = \{w : M \text{ accepts } w\}$ is the language accepted by TM M.

Observations: When M runs on input w, exactly one of the following three will happen:

a. M eventually enters an accepting state;

b. M crashes;

c. M runs forever without entering accepting state.

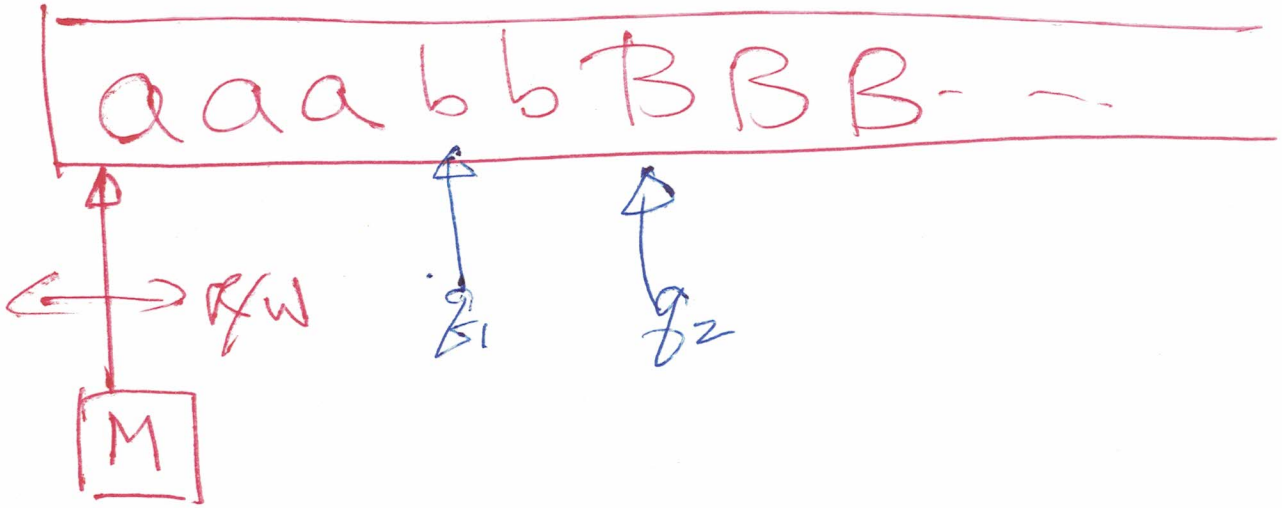
When M runs on input w, if a or b happens, we say that M halts on w.

When the TM M halts on all input, M is called an algorithm.

L is accepted by M if $L = L(M)$;

L is recognized by M if ~~let~~ $L = L(M)$ and M is an algorithm.
(decided)

Build a TM recognizing a^+b^+.



" Keep reading a's and make sure that I read at
" least one a. Keep reading b's and make sure
" that I read at least one b. When I see
" Blank, I say yes.

$$\delta(q_0, a) = (q_1, a, R)$$

$$\delta(q_1, a) = (q_1, a, R)$$

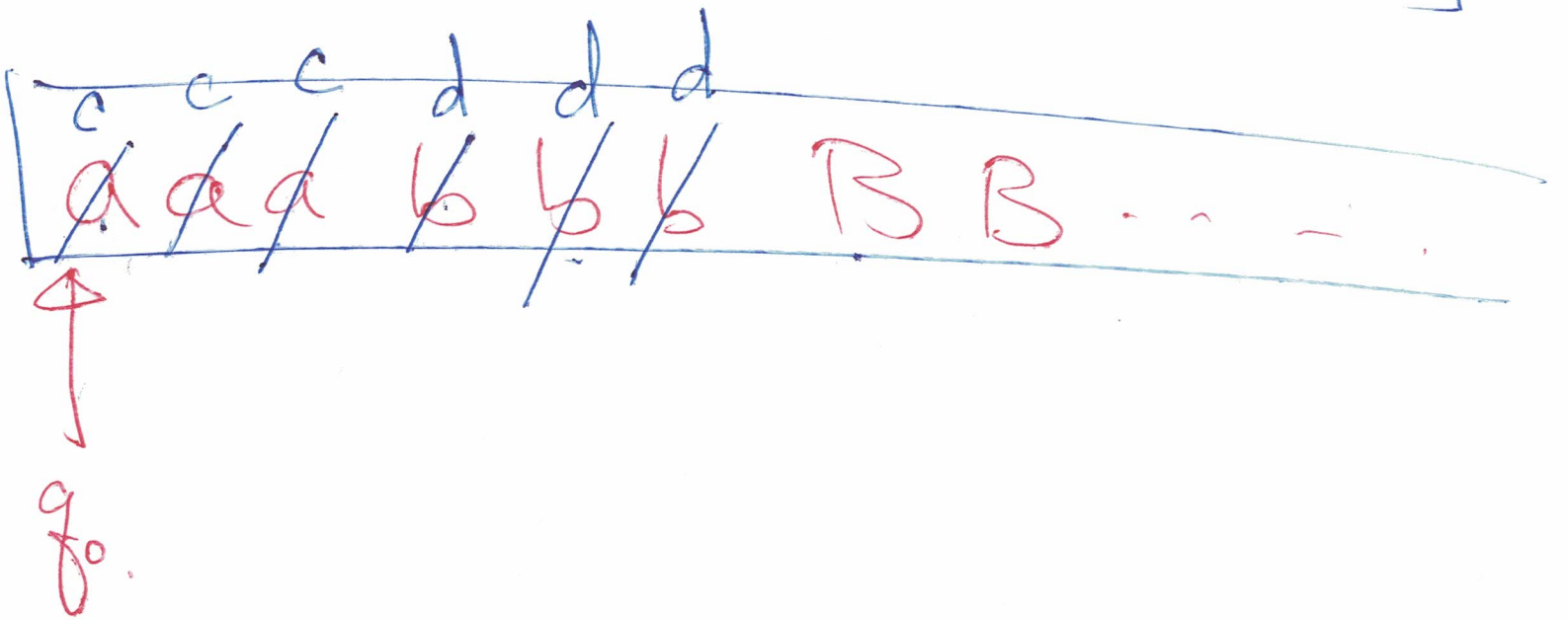
$$\delta(q_1, b) = (q_2, b, R)$$

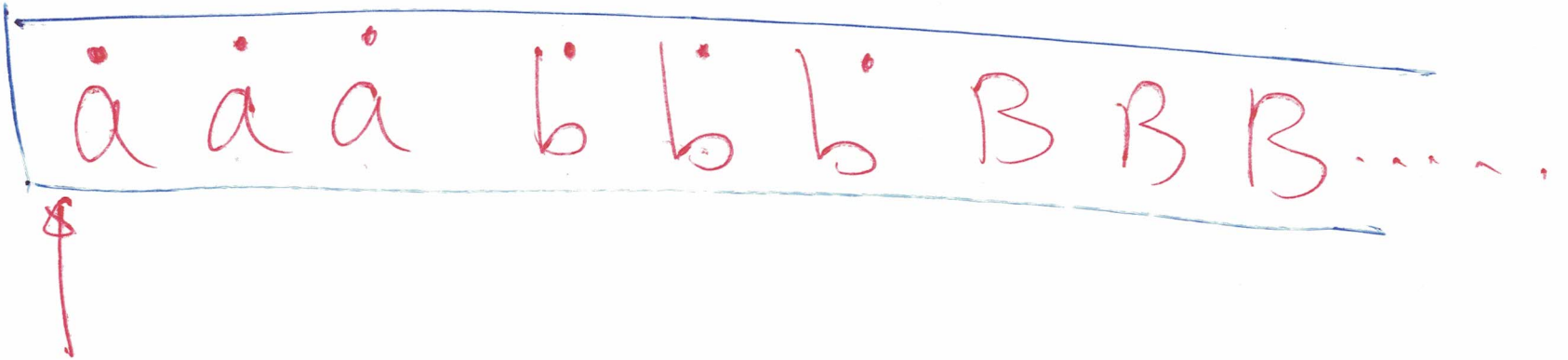
$$\delta(q_2, b) = (q_2, b, R)$$

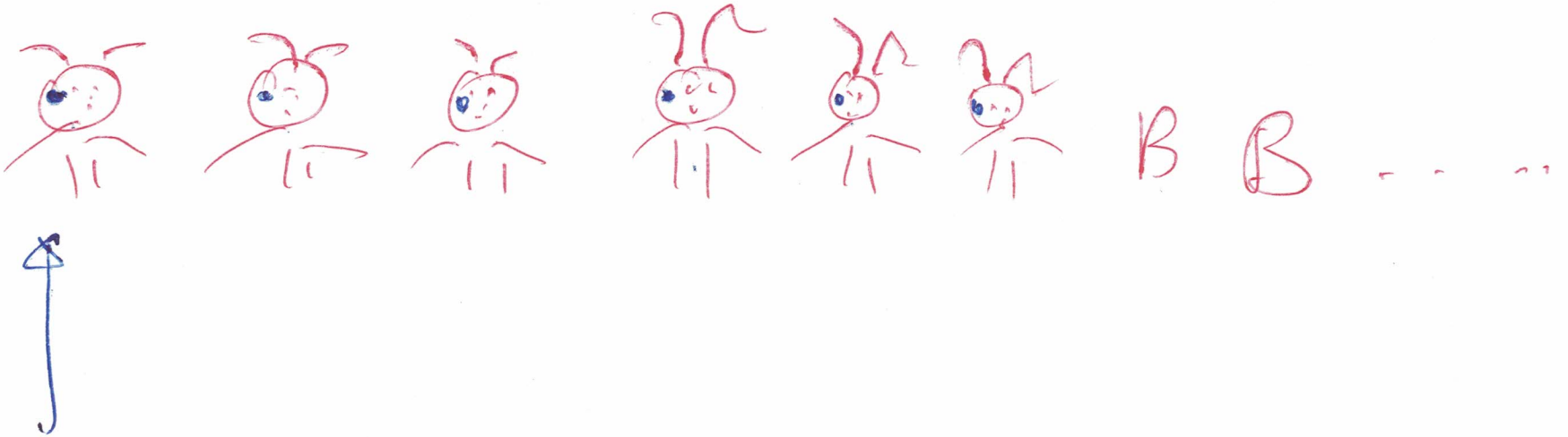
$$\delta(q_2, B) = (q_{happy}, B, S).$$

accepting state.

Build a TM recognizing $\{a^n b^n : n \geq 1\}$







next time we do coding.